
Client-Server Development

Moving from adolescence to must-have mainstream.

Client-Server Development is Like Teenage Sex

- It's on everyone's mind all the time
- Everyone is talking about it
- Everyone thinks everyone else is doing it
- The few who actually *are* doing it are
 - doing it poorly
 - sure it will be better next time
 - not practicing it safely
- Everyone is bragging about their successes, although very few have had any successes

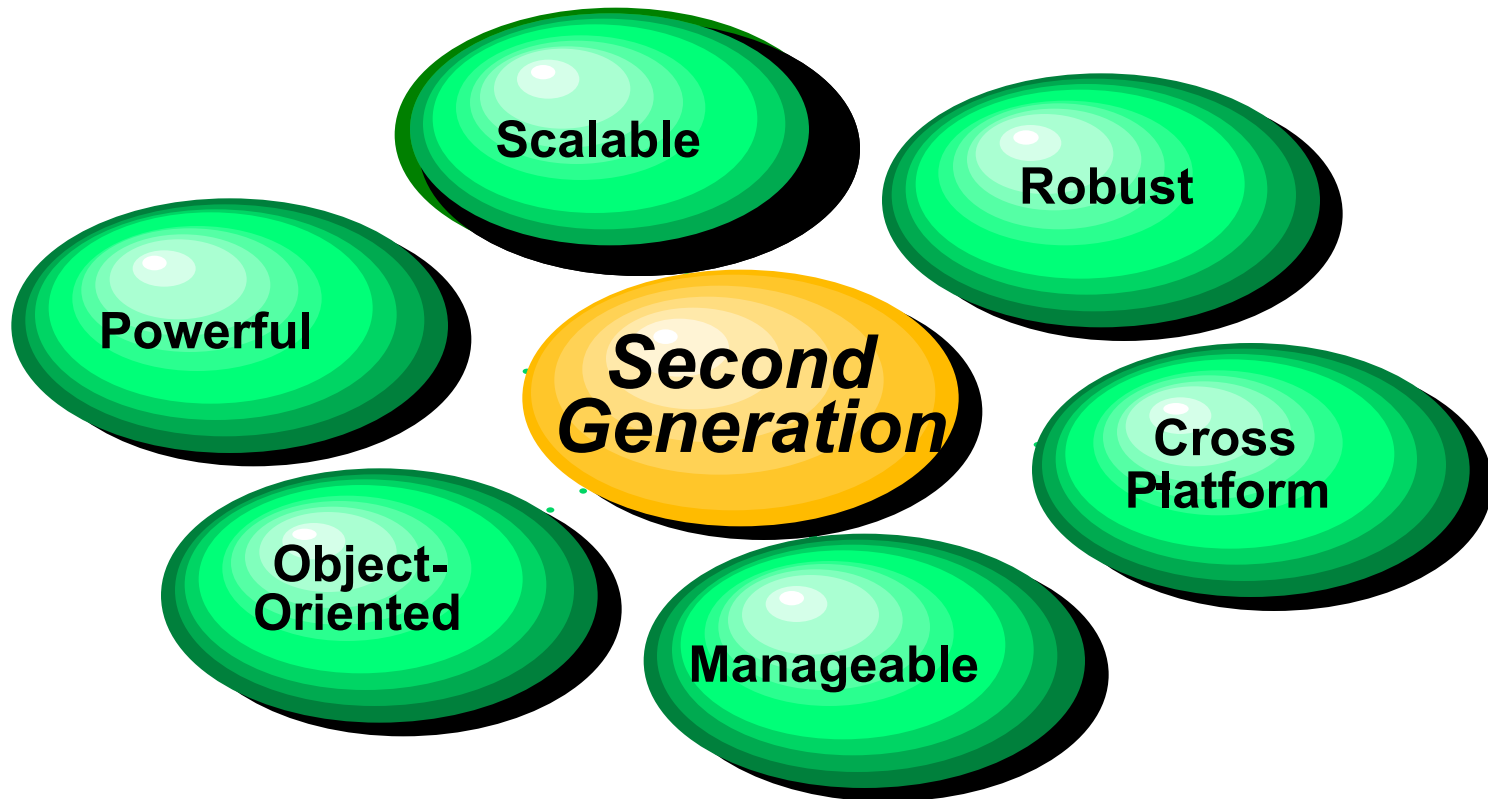
Application Development Software Market Size

- Estimate \$2-3 billion
 - Conservative definition (major suppliers competing with each other)
 - International supplier sales included
- Majority of sales in US, second most active market
Europe -- same as hardware markets
- 3GL and Client-server Application Development Environment (CADE) sales now largest
- 3GL sales partially bundled by hardware suppliers, so underestimated

Market Characteristics

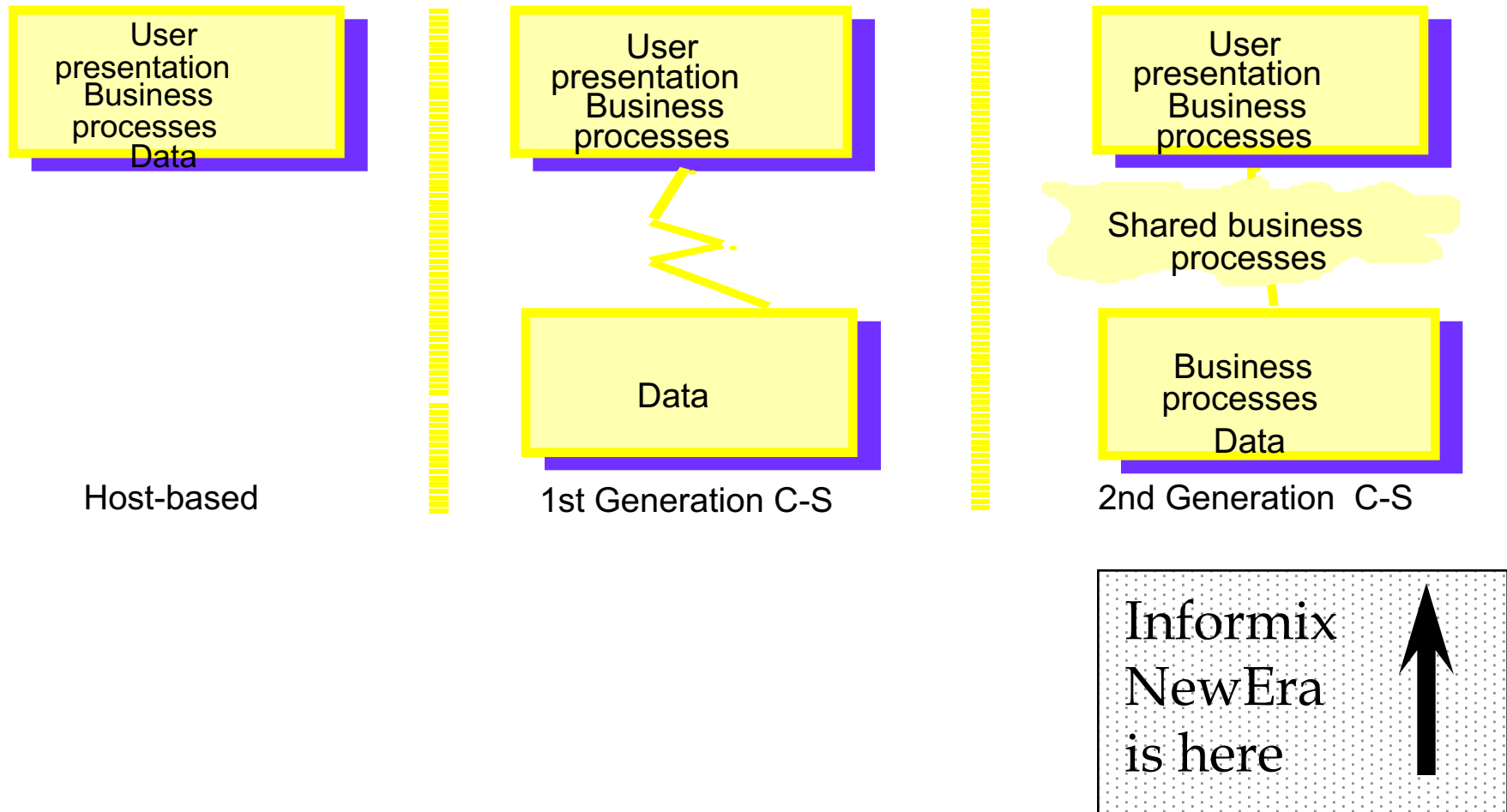
- An increasing number of application-server sales driven by development toolset, not hardware or RDBMS
- Focus on CADE market (where the action is) Key buyer criteria:
 - Scalability
 - Flexibility
 - Programmer productivity
 - Software-lifecycle support
 - Second-generation technology, especially:
 - Application partitioning
 - Business modeling
- Channels predominantly low-end but moving toward high end, direct sales
- Service emphasizes training, “learning to fish”

C-S Directions



Application Development Directions

C-S Application Architectures



Comparing Generations

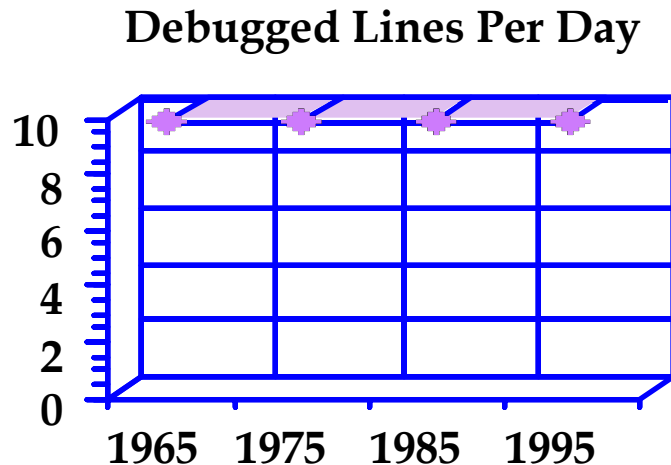
First Generation	Second Generation
• Plain Windows applications • GUI centric • Ala carte tools • Easy visual environment • Desktop & Workgroup DBMS • Fat client does all the work • Client management an afterthought • Scales under extreme duress • Speedy performance is a black art	• Distributed, cross-platform • GUI + DB + Processes • Tools <i>Family</i> • Life-cycle workbench • Unified desktop to enterprise • Partitioned, distributed • Complete systems management • Scales up and down • Performance by design & architecture

Different Tools for Different Jobs

Tool Description	Best Uses	Example
3GL	System infrastructure	C, C++
1st generation OO client-server	Workgroup decision support	Sybase PowerBuilder, Visual BASIC
Pure Object 3GL	Visual prototyping	SmallTalk, IBM VisualAge
Visual xBase	Workgroup & ad hoc apps	Microsoft FOXpro
Enterprise Application Builder	Mission critical OLTP, DSS	INFORMIX NewEra CA-OPENROAD
Object COBOL	Logic investment protection	MicroFocus COBOL CA Visual Realia

Source: Aberdeen Group

Reducing the Application Backlog



- Choose the right tool for the job!
- Give every programmer a big PC and develop on a LAN
- Use a GUI programmer's workbench
- Share the development repository

Objective is to improve on 30-year average productivity of 10 lines per day

Visual Programming Environments

- Instead of typing 3GL or 4GLcode
 - drag-and-drop or draw to create a GUI item
 - VPE automatically generates the code that goes with it
- Different VPE for every supplier, but many similar features
 - Microsoft's Visual Basic and MFC, Sybase's PowerBuilder, Forte, Oracle's Developer/2000
 - Toolsets increasingly support full software lifecycle but are usually supplemented by a 4GL
- Used for rapid development of small to medium-scale GUI desktop and client-server applications, and bundled with packaged applications
- VPEs can deliver reasonable performance and are approaching scalability to large-scale applications

User Requirements

- In the past, datacenter bought toolsets bundled with the hardware
- In desktop markets, departments and individual developers buy cheap toolsets for rapid development of small applications
- IS Buyers now emphasizing buying one server for each new application (application server)
 - Aim is often to create a competitive-advantage application
 - To customize that server, buyers buy a development toolset
- Users often forget today's leading edge tool leads to tomorrow's legacy application!

Submarkets: 3GLs

- Price varies widely
 - \$100 to \$5,000 on PC per developer
 - \$50,000 on mainframe (shared)
- IBM leads bundled suppliers (COBOL)
- Microsoft leads unbundled suppliers (C/C++ compiler)
- Channels:
 - bundled in direct sales
 - retail outlets
 - some mail-order
- Service
 - You get what you pay for
- Informix: Compiler add-ons (e.g., pre-compilers)

Submarkets: 4GLs

- **High cost (\$2000-\$50,000)**
 - Desktop license at low end
 - Mainframe licenses at high end
- **RDBMS suppliers have led**
 - Oracle
 - Informix
 - Progress
 - Ingres (pre-CA)
- **CADEs replacing 4GLs (incorporate them)**
- **Channels: bundled in direct sales, direct sales, some indirect and telemarketing**
- **Service: often part of overall service (runtime fees)**
- **Informix: Made its fame and fortune on character-based 4GL**

Submarkets: Pure Object-Oriented

- Fairly high cost (\$1000-\$10,000) per seat
- Presently, a small but emerging market
 - IBM VisualAge
 - ParcPlace PARTS
 - Developers like Microsoft MFC
- Often sold as competitor to CADEs
- Channels: telemarketing, indirect sales, some mail-order, increasing direct sales
- Service: extensive training
- Issue for Informix: Selling OO-CADE vs. pure OO to the religious zealots

Submarkets: CASE

- **High cost (\$10,000 - \$100,000)**
- **TI the market leader with Composer By IEF**
- **Second tier suppliers**
 - **Intersolv**
 - **Cadre**
 - **Knowledgeware**
- **The acronym CASE is a turn-off to many IS buyers**
- **Niche market for users with need for ongoing long-lifecycle large-scale development**
 - **Channels: mostly direct sales**
 - **Service: ongoing consulting**
- **Informix: We don't do CASE, but TI and Cadre can generate 4GL native code**

Submarkets: CADEs

- **Three cost tiers**
 - Low (free lead-in, \$400-1000)
 - Moderate (\$1000-\$3000)
 - High-end (\$3000-\$30,000)
- **Many submarkets**
 - Low-end leader Microsoft, Powersoft second
 - RDBMS-supplier leader Oracle
 - Most growing at 50%-125% per year
- **Increasingly strategic to IS buyer**
- **Channels: all types (increasing proportion of direct sales, no runtime fees at low end)**
- **Service: training, sometimes bundled with consulting and systems integration in partnership with hardware and/or RDBMS supplier**
- **Informix: NewEra is leading technology for sophisticated, long-term developers ... not a throw-away tool**

Buyer Criteria: Scalability

- **Performance scalability**
 - speed-up
 - processor scale-up
- **Development-process scalability**
 - Managing large IS projects
- **Ability to leverage RDBMS technology for scalability**
 - Very Large Data Bases
 - Many OLTP users

Buyer Criteria: Flexibility

- **Open interoperability**
 - **de facto and de jure standards**
 - **application programming interfaces**
 - **code portability**
- **Multiarchitecture portability**
- **Ability to customize toolset**
- **Ability to integrate with third-party toolsets and applications**

Buyer Criteria: Programmer Productivity

- Technologies:
 - 4GLs/scripting languages
 - Visual Programming Environments (VPEs)
 - Object-Oriented Technology

Buyer Criteria: Software-Lifecycle Support

- Support for application generation from design specs and full integration of all tools across the software lifecycle
- Support for each lifecycle stage:
 - Design
 - Coding/testing
 - Deployment
 - Maintenance

Buyer Criteria: Second-Generation Technologies

- **Application partitioning**
- **Business Modeling**
- **Other**
 - **OLTP monitors**
 - **Workflow/E-mail**
 - **Multimedia and complex data types**

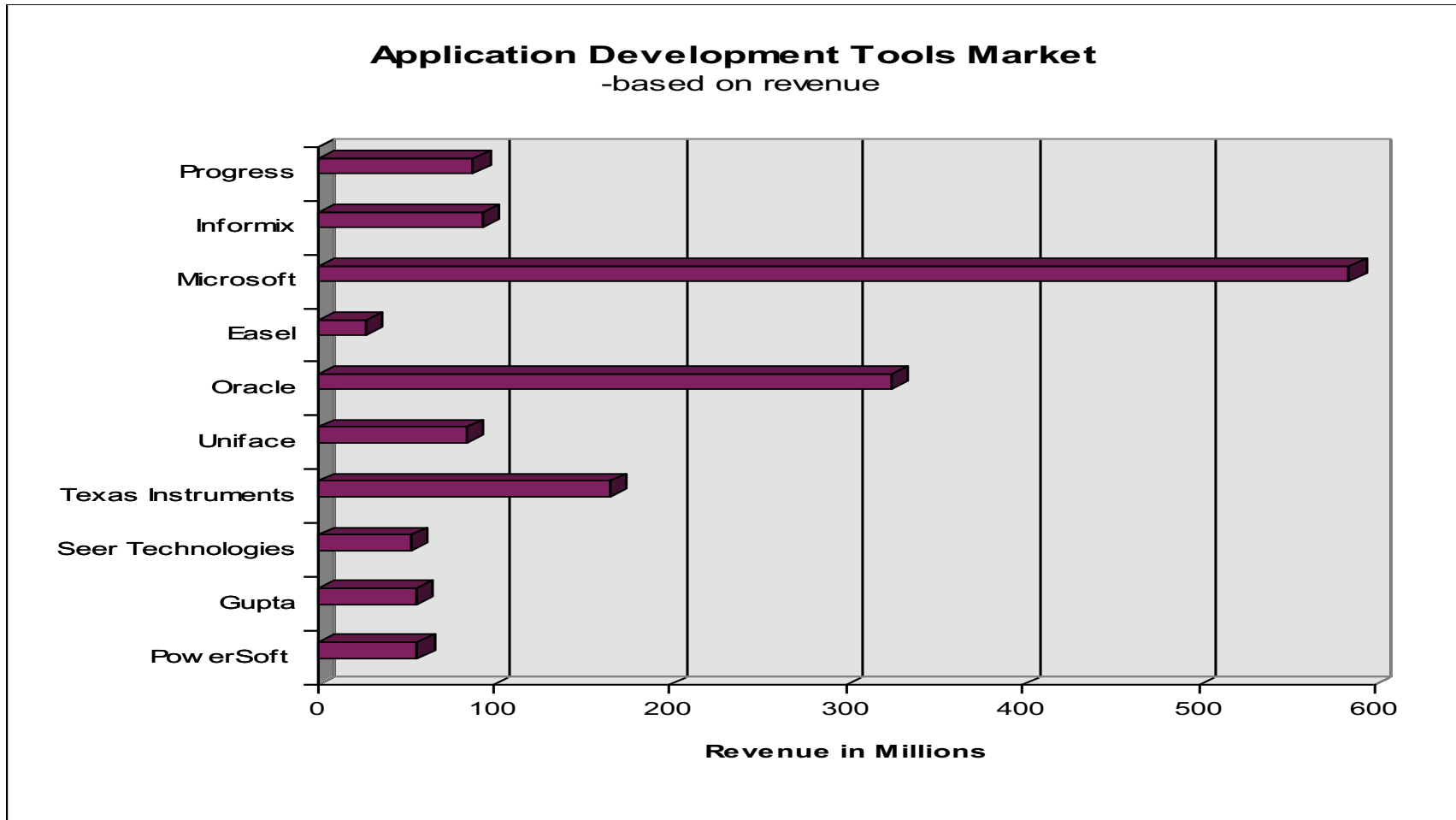
Considerations for Buyers

- **Choose a few client-server software tools -- wisely**
 - **Can the software handle your apps now and into the future?**
 - **Is the technology usable by mere-mortal programmers**
 - **Does the software integrate well**
- **Buy the software company as well as the technology**
 - **You will live with technology for 5 - 7 years**
 - **Suppliers bring the cook-book & recipes**
- **Look for integration, synergy, vision, direction, the future**

Wrap-up

- CADE is a hot market with enormous changes
- Technology is fast-moving (leapfrogging)
- Users are beginning to emphasize the long term
 - Scalability
 - Repositories
 - Second-generation technologies

Application Development Tools Market



(source: Aberdeen Group Inc..)

Supplier Profiles: Selection Criteria

- **Visibility = Perceptions of Industry Influencers**
 - Microsoft, Oracle, and Sybase/Powersoft are today's major players in revenue terms
 - TI is the major player in the CASE submarket and has attracted major attention because of its repository deal with Microsoft
 - Seer Technologies is frequently referenced-sold by IBM because of IBM's lack of client-server development technology, and is relatively large (\$100 million in revenues)
 - Forte is always cited as the second-generation technology supplier
- **Technology**
 - All of these suppliers except TI offer new client-server development technology that users perceive as valuable

Microsoft: Visual Basic

- **Focus on low end (desktop and PC LAN)**
 - **Sells ease of use, de facto standard**
 - **High visibility in IS shops**
- **Largest unit sales of any CADE supplier**
 - **Claims 4 million developers**
 - **Most channels (retail, indirect, direct)**
- **Minimal service, no runtime fees**
- **Key to the future: OLE for distributed objects**
 - **Moving slowly toward high end**
 - **Selling OLE as standard basis for CADE client (aims to own the components)**
 - **Keep Visual Basic in sync with Visual C++, Microsoft Access toolset**

Powersoft (Sybase)

- Shrink-wrap/IS developer dual market strategy
 - Sells “easy entry into client-server programming”, de facto standard due to high market penetration
 - High visibility in IS shops
- Second to Microsoft in unit sales
 - Claims approaching one million developers; 125% CAGR
 - Both indirect and direct sales ... revenue neutral comp plan
 - But Sybase and Powersoft sales forces are not integrated
- Weaknesses
 - Thoroughly tied to Windows. Not cross-platform
 - No character-terminal support
 - Poor scalability, no repository
 - Client-only; no help for *server* part of application

Oracle Developer/2000

- **Focus on high end**
 - Sell broad functionality, cross-platform support
 - Workgroup and impressive high-end Developer/2000 products, separate CASE product (Designer/2000) moved from UNIX to Windows
- **Strong installed base**
 - Doing better than Oracle Applications, but well below Oracle RDBMS sales growth rates
 - Mostly direct sales, often bundled with Oracle RDBMS and services/consulting
 - Concerned primarily with Microsoft & Sybase
- **Weaknesses**
 - Developer/2000 is new skin on old CDE toolset
 - Forms 4.5 is starting to be Windows-competitive
 - Only decent OO tool is Oracle Objects, a VB clone and incompatible with Developer/2000 tools

TI Composer by IEF

- Focus on *very* high end, CASE-oriented
 - Sells leadership in CASE market, methodology
 - New Composer CADE product hasn't yet implemented many second-generation features
 - TI-Microsoft repository joint development project in question
- Dominates mature CASE market
 - Over \$100 million in CASE revenues
 - Direct sales by dedicated reps
- Service emphasizes training in methodology
- The future:
 - Difficulty migrating from mainframe apps market to CADE market
 - No clear delivery date for repository
 - Client-server technology outpaces ability to generate apps

Seer Technologies

- Focus on high end
 - Sells broad second-generation technology with consulting/SI
 - Application partitioning
 - Business modeling
 - Mainframe-as-server emphasis
- Approaching \$100 million in revenues
 - Moving out of IBM's shelter (IBM has resold Seer, as IBM lacks a good CADE story)
 - Direct sales
- Emphasize service, including systems integration, training, support
- The future:
 - Recent IPO
 - Expanding second-generation capabilities

Fortè Software

- **Focus on high end**
 - Sell leadership in application partitioning second-generation technology
- **Start-up Company**
 - Exceptional visibility
 - \$6 million in revenues estimated in less than 2 quarters
 - Deep \$30M venture-capital pockets
 - Some joint projects with Digital
 - Direct sales model
- **Services a key source of revenue**
 - Training in how to do application partitioning
- **The future:**
 - Next release fills holes (more platforms, business modeling)

Positioning NewEra

- **NewEra deserves a long, hard look**
 - **Leading-edge OO-CADE**
 - **Full object model**
 - **Second-generation hot-buttons (partitioning, x-platform, scalability, OO, full Windows support)**
 - **Informix core and third-party class library support builds an add-on business around the tool**
- **Selling strategy**
 - **Qualify on need/desire for complex tool with long-term legs**
 - **Demand an in-depth evaluation**
 - **Go for the pilot project with a few in-house zealots**
 - **Cost issues aside, NewEra is great for Informix ISVs**